

Group 8 Project Proposal

DATS 6450.13 — Big Data Analysis at Scale

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Dataset

Source: s3://dsan6000-datasets/reddit/parquet/

Scale: ~446 GB of Reddit comments and submissions from June 2023 – July 2024, stored in Parquet format.

Contents: Post titles, bodies, comments, timestamps, subreddit metadata, author identifiers, and engagement metrics (score, upvote ratio, number of comments).

High-Level Problem Statement

What makes a Reddit post go viral — positivity, negativity, or controversy?

We analyze engagement patterns, sentiment, and content characteristics across subreddits to predict post success and understand the role emotional tone plays in driving comments, upvotes, and user participation. Our findings aim to help content creators, community moderators, and platform strategists understand which behavioral, temporal, and linguistic signals turn ordinary posts into viral conversations.

Specific Questions (10 total)

Exploratory Data Analysis (EDA)

- **Q1.** Which subreddits have the highest virality rate — i.e., the highest share of posts that cross a high-engagement threshold relative to their subreddit's average?
- **Q2.** What time of day, day of week, and posting cadence correlate with viral posts? Does posting at 9 AM EST outperform 2 AM?
- **Q3.** What is the distribution of engagement across posts? Does it follow a power law (a few posts get most of the attention), and how does that vary by subreddit?
- **Q4.** How does a user's posting history relate to virality? Do power users (frequent posters) get more engagement than first-time posters, or is it the opposite?

Natural Language Processing (NLP)

- **Q5.** How does post sentiment (positive, negative, neutral) correlate with engagement? Do negative posts actually receive more comments?
- **Q6.** What words, phrases, and content themes distinguish viral posts from average posts? We will use TF-IDF analysis to compare the vocabulary of high-engagement vs. low-engagement posts across top subreddits.
- **Q7.** Is there a “controversy signal” in the language itself? We measure sentiment variance across comments — posts where commenters strongly disagree (mixed sentiment) vs. posts where everyone agrees.

Machine Learning (ML)

- **Q8.** Can we predict the number of comments a post will receive based on title text, subreddit, posting time, and author history? (Regression)
- **Q9.** Can we classify a post as “viral” vs “not viral” using only early features (title, author, timing)? (Binary classification)
- **Q10.** Can we classify a post as “controversial” vs “consensus” based on content and sentiment features? (Binary classification, using the sentiment-variance label from Q7.)

Methodology

1. Infrastructure & Data Access

All processing will be performed on an Apache Spark cluster running on AWS EC2. The Reddit dataset (~446 GB, Parquet format in S3) will be read directly into Spark DataFrames, leveraging Parquet’s columnar format for efficient partition pruning and predicate pushdown. Given the scale, no work will be done in pandas on a single machine — all transformations, aggregations, and ML training will use PySpark and Spark MLlib.

2. Data Preparation & Feature Engineering

We will clean and preprocess submissions and comments data: filtering deleted or removed posts, handling null values, parsing timestamps into temporal features, and joining submissions with their comment threads. Derived features will include:

- Temporal features: posting hour, weekday, month, time-since-subreddit-creation
- Author features: post frequency, account age, cross-subreddit activity
- Content features: title length, body length, presence of media or links
- Engagement features: score, upvote ratio, comment count, comment velocity (comments per hour in first N hours)

3. Exploratory Data Analysis (EDA)

Using Spark SQL and DataFrame aggregations, we will answer EDA questions 1–4: virality rates across subreddits, temporal engagement patterns, engagement distributions (testing for power-law behavior), and the relationship between user posting history and virality. Results will be visualized using matplotlib and seaborn on aggregated (sampled-down) results.

4. Natural Language Processing (NLP)

All NLP work will use Spark NLP (John Snow Labs) — a production-grade library that runs natively on our Spark cluster, the same tooling used in our course NLP lab.

- Text preprocessing pipeline: We will build a Spark NLP pipeline using DocumentAssembler, Tokenizer, Normalizer, and StopWordsCleaner to clean and tokenize post titles, bodies, and comments at scale.
- Sentiment analysis: We will apply Spark NLP’s pretrained sentiment model (e.g., SentimentDLModel or the Vivekn sentiment analyzer) to classify each post and comment as positive, negative, or neutral. Sentiment distributions will then be aggregated by post, subreddit, and time window.
- Text feature extraction for classification: Using Spark NLP outputs combined with Spark MLlib’s CountVectorizer and IDF, we will generate TF-IDF features on post text for downstream ML classification (Q9 and Q10).
- Controversy signal: For each post with sufficient comment volume, we will compute the variance of sentiment scores across its comments as a quantitative measure of disagreement. High-variance posts (commenters strongly split) will be labeled

controversial; low-variance posts as consensus. This label is used both for EDA (Q7) and as ground truth for the controversy classifier (Q10).

5. Machine Learning (ML)

- Regression (Q8): Predict comment count from engineered features using Spark MLlib's Linear Regression, Random Forest Regressor, and Gradient Boosted Trees; compare using RMSE and R^2 .
- Classification (Q9, Q10): Train binary classifiers (Logistic Regression, Random Forest) to predict viral vs. not-viral and controversial vs. consensus. Evaluate using accuracy, precision, recall, F1, and ROC-AUC. Handle class imbalance via stratified sampling or class weighting.
- All models trained and evaluated on the Spark cluster with train/validation/test splits.

6. Evaluation & Iteration

We will iterate on feature engineering based on model performance, and perform feature-importance analysis to understand which signals drive virality and controversy — tying findings back to the central research question.

7. Deliverables

- Clean, documented PySpark code in a public GitHub repository
- A professional project website presenting findings, visualizations, and insights
- A final written report summarizing business recommendations and conclusions